How to Learn Python in 21 Days

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At present, Python is one of the most versatile and demanded programming languages in the IT world. Statistically, there are around **8-9 Million** Python developers across the world and the number is increasing rapidly. Meanwhile, the average salary of an **Entry-Level Python Developer** in India is around **4-6 LPA** , and its rich features such as *easier syntax* , *dynamic type* , etc. make it the most recommended programming language for beginners. Moreover, there are numerous cutting-edge technologies such as *Artificial Intelligence* , *Machine Learning* , *Big Data* , etc. that heavily rely upon Python. Indeed, if you **start to learn Python** by following the right approaches, the language has a lot more to offer you to build a successful career!!

Before moving further, let's have a brief introduction to **Python** Language. Python, designed by *Guido Van Rossum* in 1991, is a **general-purpose programming language** . The language is widely used in *Web Development* , *Data Science* , *Machine Learning* , and various other trending domains in the tech world. Moreover, Python supports multiple programming paradigms and has a huge set of libraries and tools. Also, the language offers various other key features such as *better code readability* , *vast community support* , *fewer lines of code* , and many more. Here in this article, we'll discuss a ***thorough curriculum or roadmap that you need to follow to learn Python in just 21 days!***

Roadmap to Learn Python in 21 Days

Python is one of the top learning programming languages for the past few years in a row. It is object-oriented, interpreted, and high level too. Due to its easy syntax, and readability, beginners prefer this language a lot. And once you become comfortable with the syntax, then you can move on to its applications like web development, Data Science, and machine learning. But before starting blindly, you should know How to learn Python? and for that, you can follow the below 21-day Roadmap.

1. Understand the Basics (Day: 1)

This is the first and foremost task you need to learn Python is – ***To understand the nature & basics of Python Language*** ! You're required to go through the deep *introduction* , *features & applications* of the language. Meanwhile, you're also required to **know about the installation process & setting up the path** to run Python Programs. You are also recommended to **create your first basic Python Program** that will help you to get familiar with the syntax & execution process of the Python Program. Moreover, you can explore some *Integrated Development Environments (IDEs)* also such as Pycharm, Jupyter, and various others.

- Introduction to Python Language
- Python Features and Applications
- How to Install Python?
- Hello World Program in Python
- Integrated Development Environments (IDEs) for Python

2. Learn Python DataTypes, Variables & Operators (Day: 2-3)

Now, you need to take a step forward and know about the foundational elements of Python Language – ***Variables, Datatypes & Operators*** . Although when it comes to learning Python Variables, you don't need to declare variables before using them as Python is a **Dynamically-Typed Programming Language** . Moreover, you're required to go through various **built-in Datatypes in Python** such as *Numeric* , *Boolean* , *Sequence Type* , etc. Also, you can explore Operators in Python and can practice a few basic programs as well on these topics for more clarification of the concepts.

- Operators in Python
- Operator Overloading in Python
- Python Programming Examples for Practice

3. Learn Conditional & Flow Control Statements in Python (Day: 4-5)

After going through the Python Fundamentals, Variables, Operators, etc., now you need to ***learn Python's conditionals and control flow statements*** in Python Programming. The Python Program Flow Control is concerned with various topics such as *For Loop* , *While Loop* , *Do-While Loop* , etc. Meanwhile, Conditional Statements are concerned with several concepts such as *If* , *If- else* , *Nested if-else* , etc. You're required to cover these topics in-depth and also recommended to solve several programming questions relevant to these topics. Several other topics such as Control Statements like **Break** , **Continue** , **Pass** , etc. can also be taken into consideration while learning Python.

- Decision-Making Statements in Python
- Loops in Python
- Looping Techniques in Python
- Control Statements in Python
- Chaining Comparison in Python

4. Understand String, List & Dictionary Manipulations in Python (Day: 6-7)

Once you'll get done with the above-mentioned topics, now to learn Python you're required to ***understand the String, List & Dictionary Manipulations*** in Python Language. In general, the Dictionary in Python is an unordered collection of items where each item holds a **key/value** pair. Moreover, a String in Python is an array of bytes representing Unicode characters, and a List in

Python represents a mutable data structure that concerns an ordered sequence of elements. Also, you're recommended to go through several other topics as well such as ***Tuple*** , ***Sets*** , etc.

- Python Arrays
- Python Dictionary
- Python String | Python Lists
- Byte Objects vs String in Python
- Python Sets | Python Tuples

5. Get Familiar With Python Functions & Modules (Day: 8-10)

Furthermore, you need to understand one of the most crucial parts of Python Programming – ***Functions & Modules in Python*** . You're required to learn various aspects of Python Functions such as types of functions, how to write & call a function, functions with arguments, etc. Moreover, you're also required to learn about Module in Python which is a file that contains Python definitions and statements. Also, you need to know about several other topics such as **Python Closures** , **Packages** , **Lambda functions** , and various others.

- Functions in Python
- Function with arguments
- Lambda Functions
- Python Modules | Python Package
- Python Closures

6. Go through Python File Operations (Day: 11-12)

Now, it's time to dive deeper into the world of Python Programming and ***understand File Handling and File Operations in Python*** . Indeed, Python supports file handling and allows you to handle files with various file handling options. You need to learn about various operations such as how to *Read & Write files* , *Opening & Closing files* , *Reading between the lines* , etc. To learn Python fully, you're required to go through various respective functions such as **open()** , **split()** , **append()** , and many more. You can go through with several additional topics as well such as Python seek function, etc.

- Basics of File Handling in Python
- Open a File in Python
- Reading a File | Writing to a File
- Python seek() function
- Python tell() function

7. Understand the Object-Oriented Approach in Python (Day: 13-15)

Here comes one of the most important parts of the Python learning journey – Object-Oriented Programming. You need to understand the Object-Oriented nature of Python Programming through various respective concepts such as class, object, instances, etc. You're required to learn about the OOPs paradigms such as Inheritance, Polymorphism, Encapsulation, etc. in-depth to command the language. Moreover, you need to learn about several other crucial topics as well such as Data Hiding, Object Printing, Constructors & Destructors in Python, and various others.

- Class, Object, and Members in Python
- Inheritance | Polymorphism | Encapsulation
- Data Hiding & Object Printing
- Constructors | Destructors in Python
- Garbage Collection in Python

8. Learn about Regular Expressions & Exception Handling in Python (Day: 16-18)

After covering the above-mentioned topics, now you're required to understand several more advanced and underlying Python concepts such as ***Regular Expressions*** , ***Exception Handling*** , etc. When it comes to Exception Handling while learning Python, you're recommended to cover several topics such as Errors and Exceptions in Python, User-Defined Exceptions, Python Try Except, Built-in Exceptions, etc. Moreover, you also need to focus on **Regular Expressions in Python** which signifies a sequence of characters that forms a search pattern. Several additional topics that can be taken into consideration at this stage are **Python Database Interaction** , etc.

- Python Exception Handling
- User-defined Exception | Built-in Exception
- Python Try Except
- Regular Expression in Python

- MongoDB and Python

9. Go Through Multithreading & Python CGI (Day: 19-21)

Furthermore, you need to learn about a few more advanced topics such as Multithreading, Python CGI, etc. **Multithreading in Python** is concerned with various concepts such as *Thread Control Block* , *Forking threads* , *Synchronizing threads* , etc. Meanwhile, **Common Gateway Interface (CGI) Programming** in Python concerns the set of rules that are used to establish a dynamic interaction b/w a web server and the browser. Moreover, you are also recommended to go through other crucial topics as well such as **Python Collections** , etc.

- Multithreading in Python
- CGI Programming in Python
- Python Collections
- Counters in Python
- Several Python Projects for Practice

Conclusion

So, after following the above-mentioned roadmap with the right attitude & dedication, you can easily ***command the Python Language*** . All you need to do is cover several topics each day to cover the entire curriculum in **just 21 days** . Now, what are you waiting for? Take out some time from your busy schedule for the next 21 days and dive into the Python Programming World!! But learning Python or any technology requires patience, continuity, and education and once you are comfortable with the basic syntax and concepts of Python then you can move on to its frameworks like Django and Flask to explore Web Development, also you can explore Data Analysis, Data Science, and Machine learning as well.

FAQs on Python

How to learn Python?

You have to start with the basic syntax and functions and steadily move to the advanced concepts and in between try to make some basic projects related to it, to get familiar with the syntax and also you can practice programming concepts from websites like geeksforgeeks.

What are the basics of Python?

These are some of the must-know basic concepts of Python:

1. *Variables and Data Types*
2. *Functions*
3. *loops*
4. *Input and Output*
5. *File Handline*
6. *Modules and libraries*

Can I learn web development in Python?

Yes, you can learn web development with the help of Python, using its frameworks like Django(used for the backend) and Flask (used for frontend)

Looking to dive into the world of programming or sharpen your Python skills? Our **Master Python: Complete Beginner to Advanced Course** is your ultimate guide to becoming proficient in Python. This course covers everything you need to build a solid foundation from fundamental programming concepts to advanced techniques. With **hands-on projects**, real-world examples, and expert guidance, you'll gain the confidence to tackle complex **coding challenges**. Whether you're starting from scratch or aiming to enhance your skills, this course is the perfect fit. Enroll now and master Python, the language of the future!

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How to convert DateTime to integer in Python

Python provides a module called DateTime to perform all the operations related to date and time. It has a rich set of functions used to perform almost all the operations that deal with time. It needs to be imported first to use the functions and it comes along with python, ...

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How to Build a Python package?

In this article, we will learn how to develop the package in Python. Packages are nothing but a collection of programs designed to perform a certain set of task(s). Packages are of two types, namely Built-in Packages like collection, datetime, sqlite, etc.External...

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